

TrES-5b

R-2455

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TRES-5_140604 · created 2026-08-23T00:21:49+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2456812.8493 +/- 0.0017 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.15 +/- 0.051
Transit depth (Rp/Rs)^2	2.26 +/- 1.54 [%]
Orbital Inclination (inc)	84.25 +/- 2.59
Airmass coefficient 1 (a1)	1.123 +/- 0.066
Airmass coefficient 2 (a2)	-0.074 +/- 0.059
Scatter in the residuals of the lightcurve fit is	5.88 %
Best Comparison Star	#3 - [263, 280]
Optimal Aperture	2.36
Optimal Annulus	15.47
Transit Duration (day)	0.069 +/- 0.019

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.15 ± 0.051 vs 0.142 (deviation 0.12σ)
Duration fit vs published	0.069 d vs 0.0758 d (deviation 0.27σ)
Mid-time O–C	0.2 min
Depth SNR	2.2
Residual scatter	5.88 %

Light curve

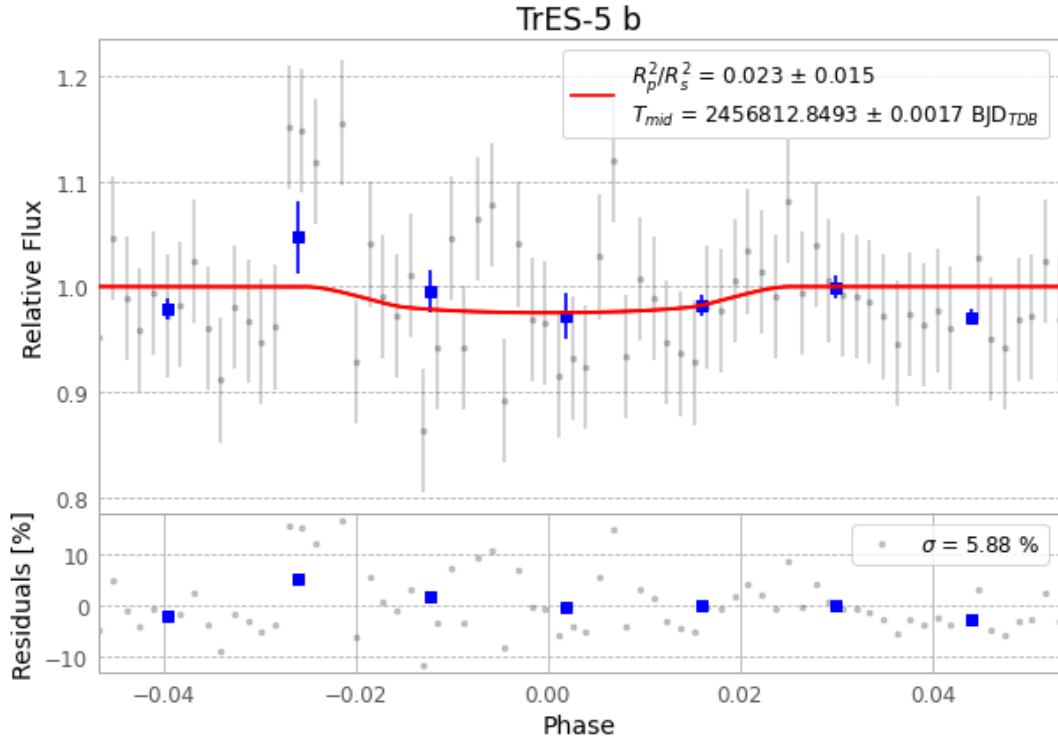


Figure 1 — FinalLightCurve_TrES-5 b_2014-06-03.png (SHA-256 756e34c5f2fd7e0f...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [357, 247], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 429fd474a4c9028f...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit a4253ce

Data provenance

72 FITS frames (47 MB), TRES-5140604064212.FITS → TRES-5140604101512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TRES-5-140604.log (ff7baf13300c...), FinalLightCurve_TrES-5 b_2014-06-03.png (756e34c5f2fd...), FinalLightCurve_TrES-5 b_2014-06-03.csv (283604b5a0ee...)

Compute resources

Wall time 128.8 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-23 00:21Z.